

PAPER

Inferring disruption of directed graphs using LIKA reveals altered protein phosphorylation networks in schizophrenia

Lujing Zhang,¹ Andrew G. Demarco,² Kimia Ghafari,³ Bernie Devlin,² Matthew L. MacDonald^{2,4} and Kathryn Roeder^{1,5,*}

¹Department of Statistics and Data Science, Carnegie Mellon University, 5000 Forbes Avenue, 15213, PA, United States of America,

²Department of Psychiatry, University of Pittsburgh School of Medicine, 3550 Terrace St, 15213, PA, United States of America, ³Center for Neuroscience, University of Pittsburgh School of Medicine, 3550 Terrace St, 15213, PA, United States of America, ⁴Biomedical Mass Spectrometry Center, University of Pittsburgh, 3550 Terrace St, 15213, PA, United States of America and ⁵Department of Computational Biology, Carnegie Mellon University, 5000 Forbes Avenue, 15213, PA, United States of America

*Corresponding author. kathryn.roeder@gmail.com

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Abstract

Motivation: Kinases regulate a multitude of protein functions, and their dysregulation is pivotal for many human diseases. Direct measurement of kinase activity, however, is often challenging; therefore, inferring activity from the behavior of their substrates is a widely adopted strategy. Nonetheless, traditional methods typically oversimplify the underlying network, ignoring that any particular substrate can be phosphorylated by multiple kinases.

Results: We present *LIKA*, a likelihood-based framework for inferring kinase activity from phosphoproteomic data. By modeling the many-to-many structure of kinase–substrate interactions, *LIKA* achieves high efficiency, even with limited data, while capturing network complexity. Simulation and cell line analyses confirm the robustness and accuracy of *LIKA*. Importantly, analysis of a phosphoproteomic dataset from schizophrenia and control subjects reveals novel dysregulated kinases.

Availability and Implementation: The implementation code and publicly available data are provided at: <https://github.com/lujingz/LIKA>.

Key words: kinase-substrate analysis, network inference, maximum likelihood inference

Introduction

Protein kinases regulate a wide range of biological processes by catalyzing the addition or removal of phosphate groups on specific substrate proteins, modulating their stability, subcellular localization, enzymatic activity, and interactions with other proteins (10). The actions of kinases have been repeatedly exploited to develop therapies against cancer, targeting individual enzymes (31). However, because kinases operate within intricate and adaptable signaling networks to modulate cellular physiology (29), understanding the architecture and dynamics of kinase signaling networks in both health and disease are crucial for effective therapeutic strategies (29).

To this end, different approaches have been proposed to evaluate kinase networks. The most widely-used one is **KSEA** (7), which infers kinase activity by assessing the enrichment of phosphorylation levels among known substrates. It aggregates signals across substrate groups and applies statistical tests to quantify pathway activation. The method

is simple and easy to implement, but it fails to account for a common case in which one or more substrates can be phosphorylated by multiple kinases. **IKAP** (21) takes into account the complexity of the network connections, but it targets the links between the kinase and phosphosites, instead of the kinases directly. Moreover, **IKAP** introduces a large number of parameters and imposes a higher requirement on the data, such as six or more time periods of measurement. **INKA** (5) integrates phosphosite and kinase information to improve the power of inference. However, it is limited to identifying the leading kinases in the network and cannot detect kinases that differ significantly in pairwise comparisons, which is necessary when aiming to identify disease-related kinases.

To illustrate the importance of accounting for shared substrates, we examine two networks with multiple kinases acting on a set of substrates (Fig. 1). Analyzing the phosphosites of each kinase in isolation can lead to misinterpretation: an unperturbed kinase could be falsely identified as dysregulated because it shares significant substrates with truly dysregulated kinases; while a dysregulated

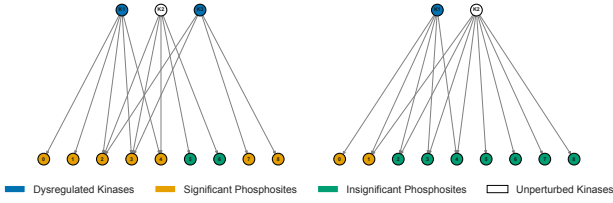


Fig. 1. Toy examples illustrating the limitation of ignoring the many-to-many nature of kinase-substrate networks. **Left:** Falsely implicating an unperturbed kinase due to dysregulated phosphosites it could influence: here, many of K2's substrates are significant solely due to the influence of K1 and K3, falsely implicating K2. **Right:** Overlooking dysregulated kinases for unperturbed phosphosites: many of K1's substrates are not significant because of the influence of K2, leading to missed detection of K1. Our method addresses these issues.

kinase may be overlooked if some of its substrates appear insignificant, for instance, due to the feedback regulation of a functional kinase. Ideally, a network analysis would explicitly account for the influence of other kinases, yielding more reliable and biologically meaningful results. We introduce *LIKA* (Likelihood Inference of Kinase Activity), a statistical framework for detecting dysregulated kinase activity that explicitly captures the many-to-many structure of kinase-substrate interactions. *LIKA* is data-efficient, computationally scalable, and capable of capturing the complexity of kinase-substrate networks. Through simulation studies, we demonstrate its robustness and accuracy. When applied to phospho-proteomic data from schizophrenia patients and from cell lines, *LIKA* successfully identifies disease-associated dysregulated kinases in the former and dominant kinases in the latter, even under conditions of limited sample sizes and highly complex networks.

Methodology

Overview of workflow

Our procedure consists of four main steps (Fig. 2). First, we collect the raw data and process them into network data and sample data (Fig. 2, Step 1). To approximate normality, the raw intensity measurements are transformed into log-intensities. If necessary (e.g., in clinical datasets), we remove the effects of clinical covariates via regression, and use the resulting residuals as the sample data. After addressing missing values, we extract the nodes that are either measured or connected to the measured nodes from a large network database, such as KSEA (7). Second, using the sample data, we obtain the substrate significant set (Fig. 2, Step 2); when the sample size is small, we employ the empirical Bayes shrinkage method (18) to improve statistical power. Third, combining the significant set with the network data, we estimate kinase dysregulation via the maximum likelihood method (Fig. 2, Step 3). Finally, we rank the kinases according to their estimated dysregulation levels (Fig. 2, Step 4).

Data Collection

The analysis requires two types of input data. First, a kinase-substrate network is used to define curated regulatory relationships derived from phosphoproteomics studies. This network is represented as a directed graph, where nodes

correspond to kinases or substrates and edges indicate kinase-substrate regulatory relationships. Second, phosphosite-level measurements from two groups of samples are used to quantify differential phosphorylation activity. To reduce potential confounding, the effects of relevant clinical covariates are adjusted for prior to downstream analysis.

Get Substrate Significant Set

Under the assumption of approximate normality, we employ per-substrate *t*-tests based on the sample mean difference and variance to obtain *p*-values. In small-sample contexts, however, as commonly encountered in cell line analyses, variance estimates for individual substrates are highly variable, which can reduce statistical reliability.

To address this issue, we employ an empirical Bayes shrinkage method (18), which borrows strength across substrates to stabilize variance estimates and thereby improve the reliability of the test statistics. Based on the resulting per-substrate *p*-values, we then apply the Benjamini-Hochberg procedure (6) to identify significant substrates, although alternative multiple testing corrections could also be considered.

Maximum Likelihood Estimation of Kinase Dysregulation

Based on biological intuition, we assume that each substrate is significantly perturbed with a probability equal to the mean dysregulation of its parent kinases, where the dysregulation levels are unknown and must be inferred from the data. To achieve this, we formulate a likelihood function in terms of kinase dysregulation and estimate the parameters using the maximum likelihood method.

Let K and S denote the sets of kinases and substrates, respectively. For $v \in K$, let $\delta(v)$ represent the dysregulation level of kinase v . For $s \in S$, let $\text{Parents}(s)$ be the set of its parent kinases. We assume that the probability of substrate s being significant is given by the mean dysregulation of its parent kinases:

$$p(s) = \frac{\sum_{v' \in \text{Parents}(s)} \delta(v')}{|\text{Parents}(s)|}.$$

Here $|\text{Parents}(s)|$ is the count of possible kinase parents or degree of the node s , which we will rewrite as D_s . Let $r(s)$ be the indicator variable denoting whether the substrate s is significant according to the significant set. The joint likelihood of the significant data $\{r(s) : s \in S\}$ is given by

$$L(\{\delta(v) : v \in K\}; \{r(s) : s \in S\}) = \prod_{s \in S} [p(s)^{r(s)} (1 - p(s))^{1-r(s)}].$$

We estimate the unknown dysregulation levels $\{\delta(v) : v \in K\}$ by maximizing this joint likelihood. Because no closed-form solution exists, the optimization is performed numerically.

Identification of Significant Kinases and Significant Subnetworks

LIKA tests whether each kinase's dysregulation exceeds the background substrate significance rate induced by the substrate-level Benjamini-Hochberg procedure. Let δ_{BH} be the realized raw *p*-value cutoff corresponding to FDR 0.05. Under the null, a substrate is called significant with probability approximately δ_{BH} , so *LIKA* uses δ_{BH} as the null dysregulation level.

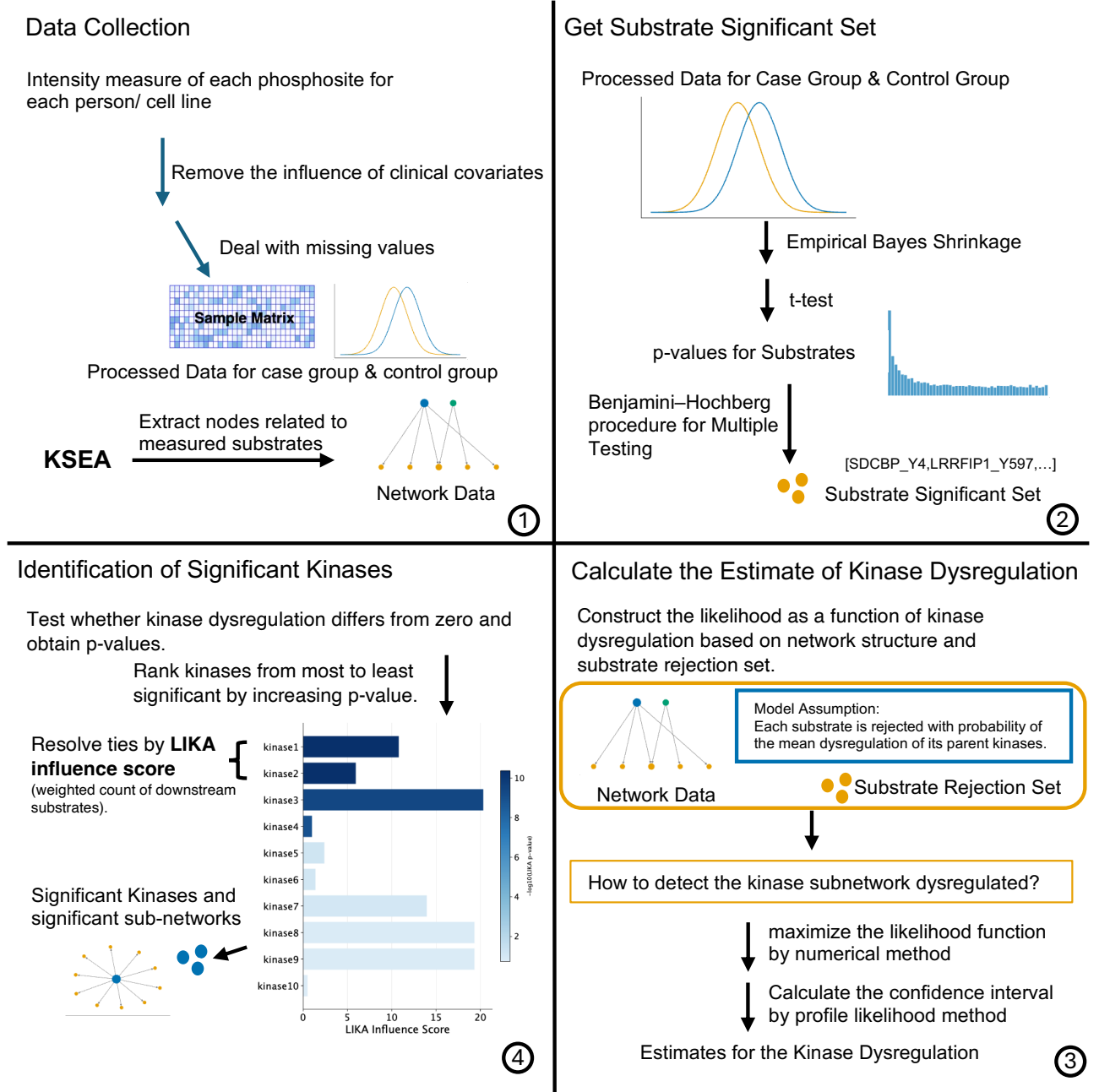


Fig. 2. Overview of LIKA workflow. (1) **Data Collection.** For substrate data, we collect and preprocess the intensity measurements of each phosphosite across individuals or cell lines. For network data, we extract nodes corresponding to the measured substrates from the KSEA dataset (7). (2) **Set of significant substrates.** When the number of samples is small (e.g., Experiment on cell line data), we apply empirical Bayes shrinkage to adjust the sample variance of each substrate, then compute p-values. After correcting for multiple testing using the Benjamini–Hochberg procedure, we obtain the set of significant substrates. (3) **Estimation of Kinase Dysregulation.** We construct a likelihood function of kinase dysregulation based on the network structure and the set of significant substrates. The likelihood is maximized numerically, and profile likelihood is used to derive confidence intervals for each kinase. (4) **Identification of Significant Kinases.** For each kinase, we test whether its dysregulation differs from zero and obtain a kinase-level p-value. Kinases are ranked from most to least significant by increasing p-value. Ties – and, in our recommendation, any set of kinases assigned very small p-values – are resolved using the LIKA influence score, defined as a weighted count of downstream substrates. The top-ranked kinases, together with their associated substrates, define the significant dysregulated subnetworks.

For kinase $v \in K$, let $\delta_{-v} = \{\delta(u) : u \in K \setminus \{v\}\}$ denote all other kinase dysregulation levels. We profile out these nuisance parameters:

$$\ell_p(d; v) = \max_{\delta_{-v}} \log L(d, \delta_{-v}; \{r(s) : s \in S\}),$$

where d is the value assigned to $\delta(v)$ and $r(s)$ indicates whether substrate s is significant. Let

$$\hat{\delta}(v) \in \arg \max_d \ell_p(d; v).$$

We test

$$H_0 : \delta(v) = \delta_{\text{BH}} \quad \text{versus} \quad H_A : \delta(v) > \delta_{\text{BH}},$$

using the profile likelihood-ratio statistic

$$T_v = 2 \left\{ \ell_p(\hat{\delta}(v); v) - \ell_p(\delta_{\text{BH}}; v) \right\}.$$

Under standard regularity conditions, $T_v \xrightarrow{d} \chi_1^2$ by Wilks' theorem (8). For the one-sided alternative, the kinase-level p-value is

$$p_v = 1 - \Phi(\text{sign}\{\hat{\delta}(v) - \delta_{\text{BH}}\} \sqrt{T_v}).$$

Kinases are ranked by increasing kinase-level p-value. For very small p-values, we additionally prioritize kinases by downstream influence. Specifically, we use the truncated p-value

$$\tilde{p}_v = \max\{p_v, \tau\},$$

where τ may be chosen as a Bonferroni-corrected significance threshold, and break ties using the LIKA influence score

$$I_{\text{LIKA}}(v) = \sum_{\{s \in S : v \in \text{Parents}(s)\}} \frac{1}{D_s}.$$

This score is the effective number of downstream substrates assigned to kinase v : substrates with fewer candidate parent kinases receive greater weight, while shared substrates are not over-counted. For each selected kinase, the significant subnetwork consists of the kinase, its significant downstream substrates, and the corresponding kinase-substrate edges.

Experiments

We conducted three sets of experiments: a simulation study; phosphorylation levels detected in cell line data, and in which our analysis contrasts levels between two cell lines; and an analysis of differential mean phosphorylation levels detected in postmortem brain samples from schizophrenia subjects versus controls. The simulation study demonstrates our model's performance on complex networks. The cell line analysis, for which ground-truth knowledge is well established, validates our model's ability to identify dysregulated kinases in real data and, importantly, to recover downstream kinases of the dominant kinase. Finally, the schizophrenia study provides biological insights into potential mechanisms underlying the disorder. The code supporting all experiments is available at <https://github.com/lujingz/LIKA>.

Simulation Results

We evaluated LIKA using simulated kinase-substrate networks with known dysregulated kinases. Substrate-level significance was generated from a binomial model in which the probability of significance depended on the dysregulation levels of the substrate's parent kinases. Significant substrates were assigned a case-control shift, with case intensities sampled from $N(2, 1)$ and control intensities from $N(0, 1)$; non-significant substrates were sampled from $N(0, 1)$ in both groups. Each group contained five samples, creating a small-sample setting for kinase inference.

We considered three network structures: non-overlapping substrates with equal kinase degree, non-overlapping substrates

with unequal kinase degree, and overlapping substrates with unequal kinase degree. Detailed network constructions are provided in Supplementary Material 1, and the simulated network structures are summarized in Supplementary Table S1. KSEA was applied to the same datasets as a baseline.

Performance was measured by precision@k and recall@k. Let G be the set of ground-truth dysregulated kinases and T_k the top k predicted kinases:

$$\text{precision@}k = \frac{|T_k \cap G|}{k}, \quad \text{recall@}k = \frac{|T_k \cap G|}{|G|}.$$

The performance across the three settings is summarized in Fig. 3. In the simplest setting, LIKA and KSEA both recovered all ground-truth kinases at the true kinase count. When kinase substrate counts were unequal, LIKA maintained perfect recovery, whereas KSEA showed reduced precision and recall. Under both degree heterogeneity and substrate overlap, LIKA consistently outperformed KSEA across a broad range of k , demonstrating greater robustness to heterogeneous and shared kinase-substrate annotations.

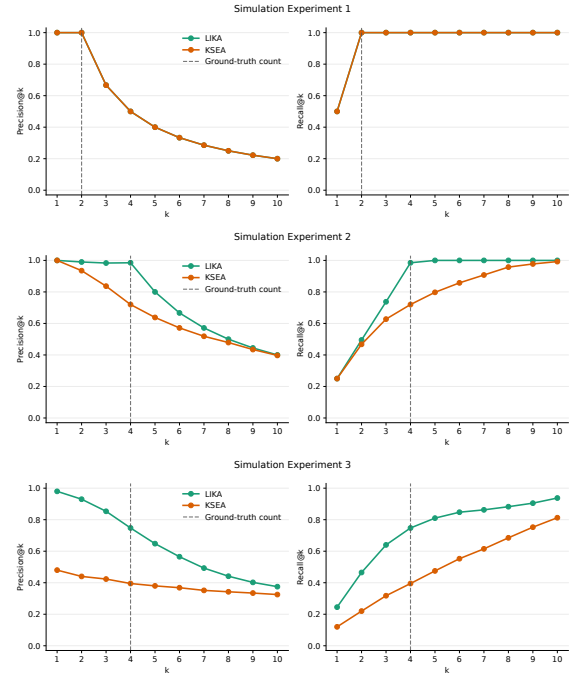


Fig. 3. Simulation performance under three synthetic kinase-substrate network structures. Precision@k (left) and recall@k (right) are averaged over 100 runs; kinases are ranked by capped LIKA p -value or KSEA p -value, and dashed lines indicate the true number of dysregulated kinases. LIKA matches KSEA in the balanced setting and outperforms it under degree heterogeneity and substrate overlap.

Experiment on cell line data

To illustrate our method when we know the ground truth, we utilize data from two cancer cell lines – K562 and HCC827-ER3 (5) – treating these cell lines as case-control samples to determine which kinases are identified by the contrast of phosphorylation levels found on proteins in the cell lines. The key mutation of K562 is a fusion of chromosomes 9 and 22, generating the fusion gene *BCR-ABL1* (27). *BCR-ABL1* substantially elevates tyrosine kinase activity, as well

as altering activity of various kinases (Serine/threonine-protein kinase CRK1, phosphoinositide 3-kinase PI3K) and pathways (RAS/RAF/MAPK, PI3K/AKT/mTOR, JAK/STAT, and WNT/-catenin) (2). The HCC827-ER3 cell line carries a heterozygous in-frame deletion of 5 amino acids (p.Glu746-Ala750del) in *EGFR*, a receptor tyrosine kinase, and also carries a homozygous in-frame mutation (p.Val218del) in *TP53*. The mutation in *EGFR* activates the Ras/mitogen-activated protein kinase and PI3K/Akt pathways (13). Although *BCR-ABL1* and *EGFR* overlap in some activation pathways, these cell lines should differ substantially in the phosphorylation patterns of their substrates.

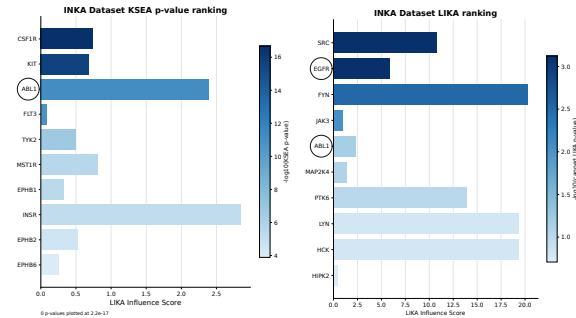


Fig. 4. Results for the INKA dataset. **Left:** Top ten kinases ranked by KSEA p-value. **Right:** Top ten kinases ranked by LIKA p-value. Kinases are ordered from rank 1 to rank 10 from top to bottom. Bar length shows the LIKA influence score, and color intensity shows $-\log_{10}(p\text{-value})$, with darker colors corresponding to smaller p-values. Ground-truth kinases reported by Beekhof et al. (5) are circled on the y-axis. LIKA identifies both EGFR and ABL1 among the top ten kinases, whereas KSEA identifies only ABL1, suggesting that LIKA better recovers the reported ground truth in this example. Complete kinase rankings and associated statistics are provided in Supplementary Table S2.

An important feature of this experiment is the sample size. Although most case-control studies have relatively large sample sizes, the data produced by Beekhof et al.(5) are limited to phosphorylation measures from two samples of each cell line, K562 and HCC827-ER3, and a technical replicate for each sample. However, because the variability of measurements over samples of the same cell line is low, we anticipate signal in these data. The two cell lines share 504 phosphosites, 296 of which are recorded in KSEA (7). We therefore limited our analyses to kinases associated with these 296 phosphosites, from which we constructed a network comprising 365 nodes. Each phosphosite has four measurements per cell line, with partial missingness. The log-transformed intensities, which are approximately normally distributed, were used as input data. An empirical Bayes method (18) was used to contrast phosphosite values. Additional preprocessing details and substrate-level results are provided in Supplementary Material 2 and Supplementary Table S3. Both EGFR and ABL1 are among the top 10 kinases detected by LIKA score (Fig. 4), while only ABL1 is detected by KSEA, highlighting the sensitivity of LIKA. In addition, beyond the two focal kinases, LIKA identifies 4 kinases that are direct downstream targets of these focal kinases, namely FYN, SRC, LYN, and HCK, whereas KSEA identifies none. Notably, both kinases regulate a large number of substrates (Fig. 5), most of which exhibit significant differences between the two cell lines. The substrates with the smallest p -values are labeled, forming an “extremely hot” region.

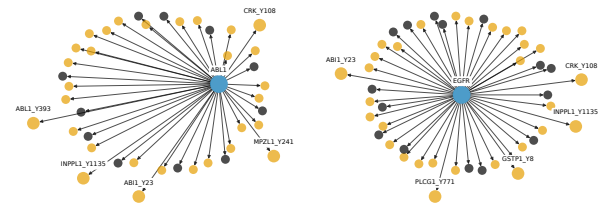


Fig. 5. Subnetworks associated with ABL1 and EGFR. Yellow nodes represent significant substrates, whereas black nodes represent insignificant substrates. Among the yellow nodes, those with labels correspond to the five most strongly significant substrates, characterized by the smallest p -values.

Perturbation of kinase function in schizophrenia

Dataset Overview

For this experiment, a postmortem sample from the dorsal anterior cingulate cortex (dACC) of each of 112 subjects was assessed by TMT-labeled mass spectrometry for phosphoproteomic detection and measurement. Of the 112, 56 of the subjects were diagnosed with schizophrenia. These subjects were pair-matched to 56 subjects who had no psychiatric diagnosis, matching on sex, age band, and postmortem interval (PMI). Pairs were then processed together for all experimental protocols. Of the 45,471 phosphopeptides detected as greater than zero abundance, the abundances were confidently estimated for 18,372. To estimate the mean phosphorylation differences between cases and controls, we first subjected these estimates to a model selection procedure in which covariates were retained if they were useful predictors for at least 5% of the phosphopeptides. The final model included case-status, as well as covariates plex (TMT batch), PMI, age, ancestry, and res.pH (residual obtained after regressing pH on diagnosis). However, of these phosphopeptides, only 1,740 could be confidently assigned to a specific phosphosite that was also recorded in KSEA (7). Thus these measurements will be the focus here. The corresponding phosphosite-level p -values, significance indicators, and upstream kinase annotations are provided in Supplementary Table S4. Of these phosphosites, 1,231 had significantly different means ($FDR < 0.05$) for cases versus controls.

Next, based on the 212 kinases associated with these 1,740 phosphosites, we constructed a network comprising 1,952 nodes. When kinases share identical substrates, it is impossible to determine which one drives the effect. To address this challenge, we merged such kinases by treating them as a single node in the network, with the union of their original parents and children assigned as the parents and children of the merged node.

Main findings

Using the kinase network and the p -values from the case-control contrast, LIKA and KSEA highlighted many of the same kinases as prominently differentiated (Fig. 6; complete kinase rankings are provided in Supplementary Table S5). Of the top 10 for each method, four overlapped, namely MAPK3, CSNK2A1, CDK1, and CDK2. Rare mutations in *CSNK2A1* have been weakly associated with schizophrenia and other neurodevelopmental disorders ((25)). The different kinases in the top 10 sets were revealing about the methods. For example, while KSEA highlighted NEK3, it had only one very significant substrate, SUGT1.T265. SUGT1.T265 was also a target for 8 other kinases, including CDK1, CDK2 and MAPK3, which were also

in the KSEA top 10 set. Thus, it is plausible that the differential case-control phosphorylation signal at SUGT1.T265 arose from the action of other top kinases, rather than NEK3. Another observation of interest: CDK1 and CDK2 shared over half of their significant substrates with MAPK3. In fact, MAPK3 (also known as ERK1) was a plausible actor for much if not all of the relevant differential phosphorylation targets, in the sense that non-coding variation impacting expression of this kinase has been implicated by early (Supplementary Information and Data file 2, (11)) and recent (Supplementary Table 12 of (30); see also (9)) colocalization of genome-wide association (GWAS) findings for schizophrenia. *MAPK3* falls in chromosome region 16p11.2, which is subject to copy number variation: duplications of this region elevate risk for schizophrenia ((20)), whereas deletions and sometimes duplications confer risk for autism spectrum disorder ((15; 32; 19)). Notably, the GWAS-discovered common risk variation also elevates expression of *MAPK3*.

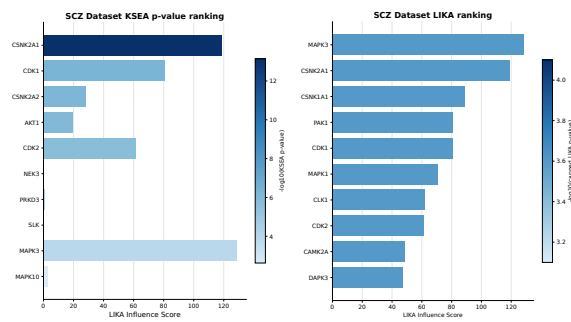


Fig. 6. Results for the schizophrenia dataset. **Left:** Top ten kinases ranked by KSEA p-value. **Right:** Top ten kinases ranked by LIKA p-value. In each panel, kinases are ordered from rank 1 to rank 10 from top to bottom. The bar length represents the LIKA influence score, while the color intensity represents the corresponding $-\log_{10}(p\text{-value})$, with darker colors indicating smaller p-values. When multiple kinases have the same p-value, ties are broken by the LIKA influence score, with the kinase having the larger score ranked higher. Complete kinase rankings and associated statistics are provided in Supplementary Table S5.

Another interesting contrast of the two methods arises for TTBK2, which showed strong association by LIKA, and SLK, which ranked highly by KSEA. Both kinases had three substrates showing differential phosphorylation. The substrates putatively phosphorylated by TTBK2, however, had no other known kinases that could phosphorylate the sites. By contrast, the substrates putatively phosphorylated by SLK could be phosphorylated by many other kinases. For this reason, LIKA reported a p-value of association of 0.40, whereas KSEA reported a highly significant p-value of association, 5.56×10^{-5} . A sensitivity analysis of the first-stage FDR threshold is provided in Supplementary Material 3.

Conclusion

Motivated by the limitations of existing kinase inference approaches, we developed LIKA, a likelihood-based framework that enables robust inference of kinase activity and dysregulation, accounting for the complexity of phospho-signaling networks. By explicitly modeling the many-to-many structure of kinase-substrate relationships, LIKA overcomes key limitations of simple enrichment-based approaches and provides a more realistic representation of biological signaling. Our

results across simulation, cell line, and schizophrenia datasets demonstrate that incorporating network structure into statistical inference substantially improves both the accuracy and interpretability of kinase activity estimates.

One of the main strengths of LIKA is its ability to disentangle overlapping kinase-substrate relationships, a pervasive feature of cellular signaling systems. Protein phosphorylation networks are inherently redundant and highly interconnected with individual phosphosites often regulated by multiple kinases depending on cellular environment (16). Traditional approaches such as KSEA implicitly assume independence among substrates, which can lead to systematic biases when substrates are shared across kinases. By contrast, LIKA formulates kinase inference as a global optimization problem, which allows to partition shared substrate signals across multiple candidate kinases. This results in improved robustness in complex network settings, as demonstrated in our simulation studies.

Our application to cell line data highlights LIKA's ability to identify not only primary driver kinases but also downstream signaling components. In the comparison between K562 and HCC827 cell lines, LIKA identified both the oncogenic drivers (BCR-ABL1 and EGFR), along with several Src-family kinases (FYN, SRC, LYN, and HCK). This observation is consistent with extensive literature demonstrating that oncogenic kinases propagate signals through downstream kinase cascades, particularly via Src-family kinases, which serve as central hubs in tyrosine kinase signaling (23; 33). This successful identification of well documented downstream effectors indicates that LIKA captures not only first-mover kinase but also higher-order network propagation effects, critical for understanding pathway-level dysregulations.

In our analyses of the schizophrenia versus control differential phosphorylation, both LIKA and KSEA converge on four kinases in their top 10, namely MAPK3, CSNK2A1, CDK1, and CDK2. P-values for the LIKA top ten are uniformly small, and therefore we rank kinases by the LIKA influence score. For KSEA, p-values for these kinases are also small, the largest being MAPK10 at p-value = 0.0024. Nonetheless, the rankings with these kinases can be very different; LIKA places MAPK3 first by LIKA influence score (Fig. 6); whereas KSEA ranks it in the ninth position by p-value (Supplementary Table S5). In either case, MAPK3 is of keen interest because genetic studies have implicated increased MAPK3 expression in schizophrenia risk, including evidence from 16.p11.2 duplication studies and transcriptomic analyses prioritizing MAPK3 as a candidate susceptibility gene ((11; 30; 20)). For this reason, we asked whether potential phosphosites targeted by MAPK3 show greater levels of phosphorylation in SCZ than controls, on average? The distribution (Supplementary Fig. S1A) of differential phosphorylation MAPK3 phosphosites shows a long right tail, consistent with genetic evidence implicating MAPK3-associated signaling in schizophrenia. Nonetheless, and intriguingly, both tails are heavier than expected, suggesting MAPK3-associated signaling in brain of individuals with schizophrenia is more complex than what would be predicted from the genetic association alone and may reflect substrate-specific or context-dependent remodeling of ERK signaling.

While its association with schizophrenia is not established with certainty ((25; 28)), CSNK2A1 has profound impact on neurodevelopment. Mutation of one copy of the gene can give rise to intellectual disability, behavioral problems, speech problems, and microcephaly, among other developmental

challenges ((22)). Such mutations are also associated with autism spectrum disorder ((12)), which is genetically correlated with schizophrenia ((3; 26; 14)). CSNK2A1 is a component of the catalytic domain of casein kinase 2 (CK2), as is CSNK2A2. CSNK2A2 is scored highly by LIKA and KSEA: LIKA yielding a p-value of 1.1×10^{-16} ; KSEA yielding a p-value of 5.9×10^{-7} . In fact, CK2 is a tetramer consisting of two active subunits, either homo- or heterodimers of these two proteins, with CSNK2B serving as the two regulatory subunits ((17)). That both active domain subunits are implicated in schizophrenia by LIKA and KSEA is therefore intriguing: perhaps CK2, when composed of the heterodimer active domain, plays an important regulatory role affecting development of schizophrenia. Contrary to MAPK3, the phosphosites for both CSNK2A1 and CSNK2A2 tend to be uniformly dephosphorylated in subjects with schizophrenia (Supplementary Fig. S1B-C). Postmortem studies have reported reduced CK2 levels and activity in cortex in subjects with schizophrenia, including reduced phosphorylation of the CK2 substrate syntaxin-1 ((4); (1)), making the tendency toward lower phosphorylation of predicted CK2 substrates remarkably consistent with prior observations.

Notably, CK2 and ERK1 are mechanistically linked. CK2 regulates the nuclear translocation and activity of ERK1 and ERK2 through phosphorylation of key serine residues required for importin-7 binding ((24)). ERK2 (MAPK1) was also among the top kinases identified by LIKA. The simultaneous implication of MAPK3, MAPK1, and both CK2 catalytic subunits therefore raises the possibility that these findings reflect perturbations within an interconnected signaling network rather than independent kinase abnormalities.

From a methodological standpoint, the stability of LIKA across a wide range of hyperparameter choices further underscores its robustness. The observed consistency in kinase rankings across varying FDR thresholds and confidence interval lengths suggests that the inferred signaling patterns reflect underlying biological structure rather than artifacts of statistical tuning. This robustness is particularly important in high-dimensional phosphoproteomics data, where arbitrary thresholding can substantially impact downstream interpretation.

Overall, LIKA provides a statistically rigorous and biologically informed framework for kinase inference in complex signaling networks. By integrating network topology with probabilistic modeling, it enables more accurate identification of dysregulated kinases and their associated pathways. Its application to schizophrenia data links genetic variation affecting MAPK3 (ERK1) to its impact on differential phosphorylation in brain tissue, offering a more integrated view of disease-related signaling alterations. More broadly, LIKA represents a generalizable approach for dissecting signaling network perturbations across diverse biological contexts.

Competing interests

No competing interest is declared.

Data and Code Availability

The source code, documentation, and scripts required to reproduce the analyses are available at <https://github.com/lujingz/LIKA>. The repository includes the implementation of LIKA, scripts for the simulation studies, and scripts for

reproducing the cell-line and schizophrenia analyses described in this article.

The kinase-substrate network used in this study was obtained from the KSEA web resource <https://casecpb.shinyapps.io/ksea/>. The cell-line phosphoproteomic data were obtained from Supplementary Dataset EV2 of Beekhof et al. (5). Processed substrate-level statistics, kinase-substrate annotations, LIKA rankings, KSEA rankings, and supplementary results are provided in Supplementary Tables S2–S5.

The schizophrenia phosphoproteomic data analyzed in this study are available [insert availability statement, accession number, or controlled-access information]. All processed data necessary to reproduce the reported kinase-level analyses are provided in the supplementary materials and/or the GitHub repository.

Author contributions statement

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Supplementary Material 1: Simulation Details

This section provides additional details on the simulated kinase–substrate networks used to evaluate the performance of the proposed method. The simulations were designed to assess whether the method can recover dysregulated kinases under different network topologies, including settings with balanced kinase degrees, heterogeneous kinase degrees, and overlapping substrate annotations. Each simulated dataset contains two groups: a control group and a case group. For every substrate, five control intensities and five case intensities were generated. Control samples were always drawn from a standard normal distribution, whereas case samples were shifted only for substrates selected as perturbed by the dysregulated kinases.

Kinase–substrate networks were represented as directed bipartite graphs. Each edge was directed from a kinase to a substrate, so an edge $K_i \rightarrow s_j$ indicates that kinase K_i regulates substrate s_j . The input network table therefore contains one row per kinase–substrate edge, with a kinase identifier in the `from` column and a substrate identifier in the `to` column. Kinases were named $K0, K1, \dots$, and substrates were named by integer identifiers. In the first two simulations, substrate sets were non-overlapping, so every substrate had exactly one upstream kinase. In the third simulation, substrate sets were allowed to overlap, so a substrate could have multiple upstream kinases.

We considered three simulated network structures:

- **Simulation 1: non-overlapping network with equal kinase degree.** This network contains 10 kinases, named $K0$ to $K9$. Each kinase is connected to 10 distinct substrates, resulting in 100 substrates in total. The substrate sets are non-overlapping: $K0$ regulates substrates 0–9, $K1$ regulates substrates 10–19, and so on. Thus, each substrate is regulated by exactly one kinase. We set $K0$ and $K1$ as the dysregulated kinases. Accordingly, substrates 0–19 were perturbed in the case group, whereas substrates 20–99 were not perturbed.
- **Simulation 2: non-overlapping network with unequal kinase degree.** This network contains 13 kinases, named $K0$ to $K12$, with heterogeneous numbers of downstream substrates. The kinase substrate counts are

20, 20, 15, 15, 12, 11, 10, 8, 6, 4, 3, 2, 1,

respectively, resulting in 127 distinct substrates in total. Substrates are assigned consecutively to kinases without overlap. For example, $K0$ is connected to the first 20 substrates, $K1$ to the next 20 substrates, and so forth. We set $K0$, $K2$, $K5$, and $K8$ as the dysregulated kinases. Because every substrate has exactly one upstream kinase in this simulation, all substrates downstream of these dysregulated kinases were perturbed, while substrates downstream only of non-dysregulated kinases were not perturbed.

- **Simulation 3: overlapping network with unequal kinase degree.** This network also contains 13 kinases, named $K0$ to $K12$, with the same kinase substrate counts used in Simulation 2:

20, 20, 15, 15, 12, 11, 10, 8, 6, 4, 3, 2, 1.

In contrast to the first two simulations, all kinase substrate sets were sampled from a shared pool of 40 substrates. For each kinase, the specified number of substrates was sampled without replacement from this pool. Because different kinases sampled from the same substrate pool, the same substrate could be regulated by multiple kinases. This setting represents the many-to-many regulatory relationships commonly observed in kinase–substrate networks. We again set $K0$, $K2$, $K5$, and $K8$ as the dysregulated kinases.

Generation of perturbed substrates

After constructing each network, we determined which substrates were perturbed in the case group. Let $\mathcal{P}(s)$ denote the set of upstream kinases for substrate s , and let $\mathcal{D}$ denote the set of dysregulated kinases. For Simulations 2 and 3, the probability that substrate s was perturbed was defined as

$$\Pr(s \text{ is perturbed}) = \frac{|\mathcal{P}(s) \cap \mathcal{D}|}{|\mathcal{P}(s)|}.$$

This rule has a simple interpretation. If all upstream kinases of a substrate are dysregulated, the substrate is always perturbed. If none of its upstream kinases are dysregulated, the substrate is never perturbed. If a substrate is shared by both dysregulated and non-dysregulated kinases, the perturbation probability is proportional to the fraction of its upstream kinases that are dysregulated. This construction aligns with the modeling assumption used by LIKA: the observed significance status of a substrate is treated as arising from the combined influence of its upstream kinases. In particular, when a substrate has multiple candidate upstream kinases, its perturbation probability depends on how many of those upstream kinases are truly dysregulated, rather than assigning the substrate exclusively to a single kinase. Therefore, the simulation directly tests whether LIKA can use shared substrate information to recover the underlying dysregulated kinases in a setting consistent with its many-to-one kinase–substrate likelihood model.

In Simulation 2, every substrate has exactly one upstream kinase, so this rule reduces to a deterministic assignment: substrates downstream of $K0$, $K2$, $K5$, or $K8$ are perturbed with probability 1, and all other substrates are perturbed with probability 0. In Simulation 3, substrates may have multiple upstream kinases, so perturbation assignment is stochastic for substrates shared by both dysregulated and non-dysregulated kinases. This design makes Simulation 3 more challenging because a perturbed substrate may be connected to several candidate kinases, not only to the true dysregulated kinase.

Simulation 1 used a deterministic perturbation design. Since $K0$ and $K1$ were specified as dysregulated and each has 10 non-overlapping substrates, substrates 0–19 were perturbed and substrates 20–99 were unperturbed.

Generation of case and control intensity data

For each substrate, we generated two rows in the simulated intensity matrix: one row for the control group and one row for the case group. Each row contains five replicate intensity measurements. The control group was generated as

$$X_{s,r}^{(0)} \sim \mathcal{N}(0, 1), \quad r = 1, \dots, 5,$$

for every substrate s . The case group was generated according to whether the substrate was selected as perturbed:

$$X_{s,r}^{(1)} \sim \begin{cases} \mathcal{N}(2, 1), & \text{if substrate } s \text{ is perturbed,} \\ \mathcal{N}(0, 1), & \text{otherwise.} \end{cases}$$

Thus, perturbed substrates have an expected case-control shift of 2, whereas unperturbed substrates have no expected difference between the two groups. No log transformation was applied to the simulated intensities when running LIKA or KSEA.

Repeated simulation runs and evaluation

For the repeated simulation evaluation, each experiment was run 100 times using different random seeds. The random seed controls the generated control and case intensities in all simulations. In Simulations 2 and 3, it also controls the stochastic selection of perturbed substrates. In Simulation 3, it additionally controls the random assignment of substrates to kinases, because the overlapping network is generated by sampling substrate sets from a shared pool.

For each run, both LIKA and KSEA were applied to the same simulated network and intensity matrix. LIKA candidates were ranked by kinase-level LIKA p -value, with ties broken by the number of efficient substrates. KSEA candidates were ranked by increasing KSEA p -value. For each method, the top k predicted kinases were compared against the known dysregulated kinase set. Let P_k denote the set of top k predicted kinases and let D denote the set of true dysregulated kinases. Precision@ k was defined as

$$\text{Precision@}k = \frac{|P_k \cap D|}{|P_k|}.$$

This quantity measures the fraction of selected candidates that are true dysregulated kinases. A high precision@ k indicates that the top-ranked list has few false positives. Recall@ k was defined as

$$\text{Recall@}k = \frac{|P_k \cap D|}{|D|}.$$

This quantity measures the fraction of all true dysregulated kinases recovered among the top k predictions. A high recall@ k indicates that the method places most or all true dysregulated kinases near the top of the ranking. Precision and recall emphasize complementary aspects of ranking quality: precision is most informative about the reliability of a short candidate list, whereas recall is most informative about whether the full ground-truth signal has been captured by a given cutoff. We calculated both metrics for $k = 1, \dots, 10$ and averaged them across the 100 simulation runs.

Table S1 summarizes the number of kinases, the number of distinct substrates, the dysregulated kinases, and whether substrate overlap is present in each simulated network.

Simulation setting	Number of kinases	Number of distinct substrates	Substrate overlap	Dysregulated kinases
Simulation 1: non-overlapping, equal-size	10	100	No	$K0, K1$
Simulation 2: non-overlapping, unequal-size	13	127	No	$K0, K2, K5, K8$
Simulation 3: overlapping, unequal-size	13	40	Yes	$K0, K2, K5, K8$

Table S1. Summary of the simulated kinase-substrate network structures. Simulation 1 evaluates performance under a balanced network topology in which all kinases have the same number of non-overlapping substrates. Simulation 2 introduces heterogeneity in the number of substrates regulated by each kinase while preserving non-overlapping substrate sets. Simulation 3 uses the same heterogeneous kinase degrees as Simulation 2 but samples downstream substrates from a shared substrate pool, allowing substrates to be regulated by multiple kinases.

Supplementary Material 2: Cell-line Experiment Details

For the substrate-intensity data, we downloaded Supplementary Dataset EV2 from Beekhof et al.(5) For kinase-substrate relationships, we used the network distributed with the KSEA method, downloaded from <https://casecpb.shinyapps.io/ksea/>. We excluded phosphosites that could not be mapped to the KSEA network and phosphosites that were not observed in either of the two cell lines analyzed here, K562 and HCC827-ER3.

Because not more than four samples were available for each cell line, substrate-specific variance estimates from ordinary t -tests are expected to be noisy and sensitive to outlying or unusually small variance estimates. To improve the stability of the substrate-level inference, we therefore used the variance adaptive shrinkage (VASH) empirical Bayes method (18). VASH borrows information across all retained substrates to estimate the overall distribution of variances, while still allowing each substrate to retain its own variance estimate.

For substrate j , let $\hat{s}_j^2$ denote the estimated variance associated with its effect estimate, s_j^2 the corresponding unknown true variance, and d_j the residual degrees of freedom. VASH uses the standard sampling model

$$\hat{s}_j^2 \mid s_j^2 \sim s_j^2 \frac{\chi_{d_j}^2}{d_j}.$$

The true variances across substrates are assigned a flexible unimodal empirical Bayes prior, represented as a mixture of inverse-gamma distributions constrained to share a common mode. The prior mode and mixture proportions are estimated from the observed variance estimates across all retained substrates by maximizing the marginal likelihood, so the degree of shrinkage is learned from the data rather than fixed in advance.

For each substrate, the observed variance estimate is combined with the fitted cross-substrate prior to obtain a posterior mixture of inverse-gamma distributions. Following Lu and Stephens (18), we used the inverse of the posterior mean precision,

$$\tilde{s}_j^2 = \left\{ \mathbb{E} \left(s_j^{-2} \mid \hat{s}_j^2 \right) \right\}^{-1},$$

as the moderated variance estimate. Substrate-level two-sided p -values were then calculated using the corresponding VASH moderated t -test. This empirical Bayes moderation reduces the influence of unstable raw variance estimates, particularly spuriously small estimates that can arise in small-sample settings, while preserving substrate-specific heterogeneity.

The resulting substrate-level p -values and corresponding network annotations are provided in the INKA phosphosite supplementary table described in the main text.

Supplementary Material 3: SCZ Substrate Distributions and Sensitivity Analysis

Phosphosite list

The list of phosphosites was generated from phosphopeptide data, using in house R scripts adapted from Wozniak et al. (40). Phosphosites were only retained with a localization score of > 0.75 and mapped to their corresponding position with the protein using human reference proteome UP000005640 (34), as a reference. To map phosphosites to kinase/substrate interactions, KSEA (39) was utilized, with databases (Phosphosite (36) PLUS, NetworKIN (37)), providing a final list of kinase/phosphosites.

Differential phosphorylation

Supplementary Fig. S1 shows the schizophrenia case-control differential phosphorylation distributions for phosphosites downstream of MAPK3, CSNK2A1, and CSNK2A2, as discussed in the main text.

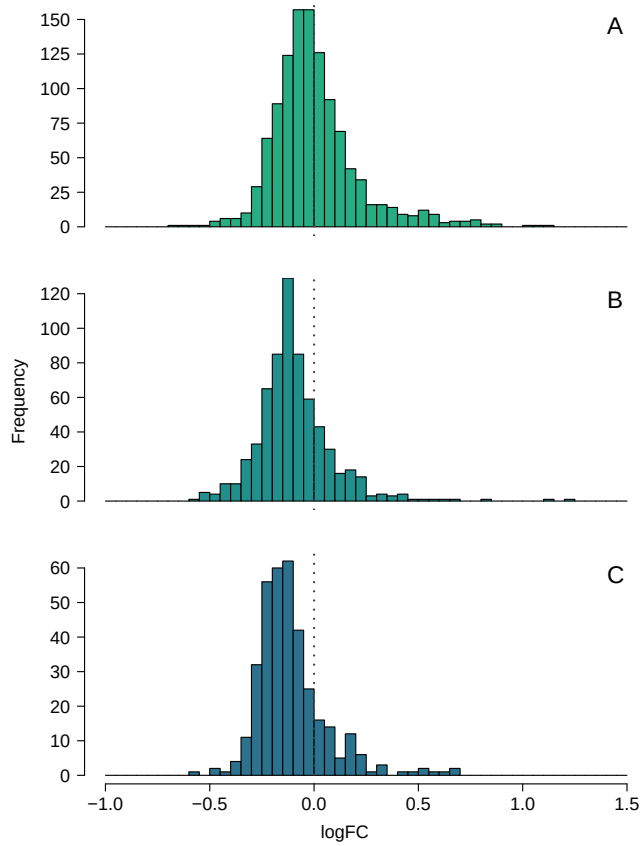


Fig. S1. Distribution of differential phosphorylation among predicted downstream phosphosites for MAPK3, CSNK2A1, and CSNK2A2 in the schizophrenia dataset. Panel a shows MAPK3-associated phosphosites, and panels b–c show phosphosites associated with the CK2 catalytic subunits CSNK2A1 and CSNK2A2. The MAPK3 distribution shows a long right tail, whereas the CK2-associated phosphosites tend to be shifted toward lower phosphorylation in schizophrenia cases.

Sensitivity analysis for LIKA

We assessed whether LIKA kinase rankings are sensitive to the first-stage substrate-level FDR threshold. In the LIKA workflow, empirical Bayes phosphosite p -values are thresholded using the Benjamini–Hochberg procedure to define the significant phosphosite set used for kinase-level inference. A robust ranking should therefore remain stable across reasonable FDR choices.

For the INKA and SCZ datasets, we repeated LIKA using first-stage FDR levels in $\{0.01, 0.025, 0.05, 0.10, 0.20\}$.

At each threshold, we recomputed the significant phosphosite set and used the largest raw substrate p -value among rejected phosphosites as the fixed null probability in the LIKA kinase-level test. Kinases were ranked by capped LIKA p -value, defined as $\max(p, 0.05/m)$, where m is the number of kinases with valid LIKA p -values; ties were broken by the number of efficient substrates. We compared rankings across FDR levels using the top-10 overlap ratio, $\frac{|T_a \cap T_b|}{10}$, where T_a and T_b denote the top-10 kinase sets from two FDR thresholds.

Supplementary Fig. S2 shows the resulting top-10 overlap ratios across FDR thresholds.

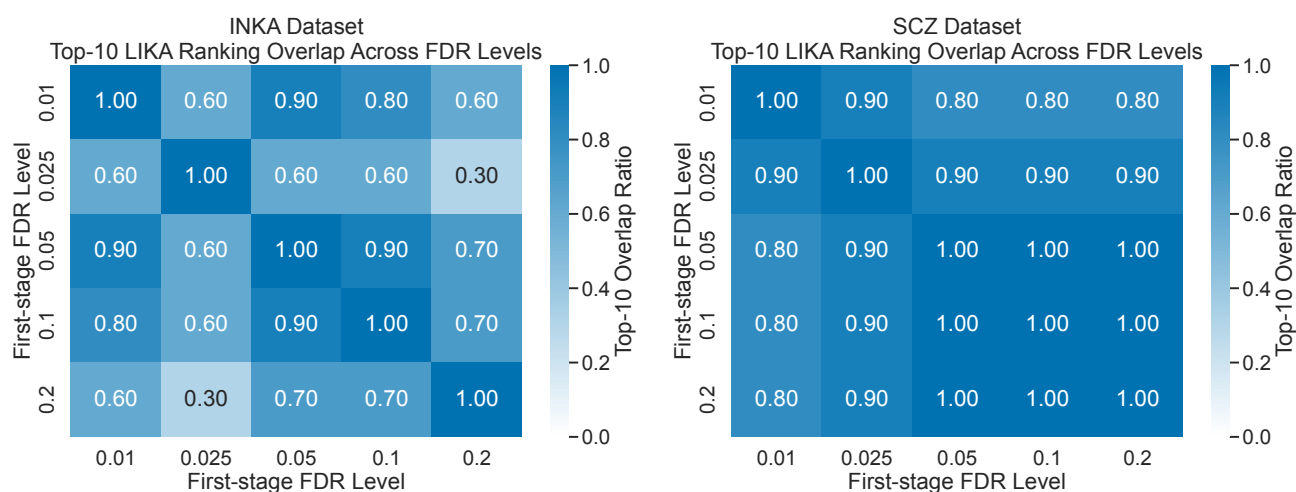


Fig. S2. Sensitivity of LIKA top-10 kinase rankings to the first-stage FDR threshold. Each heatmap entry shows the top-10 overlap ratio, $|T_a \cap T_b|/10$, between two FDR settings. The left panel shows the INKA dataset and the right panel shows the SCZ dataset; darker blue indicates stronger agreement. SCZ rankings were highly stable, with all pairwise overlaps at least 0.80 and complete overlap among FDR 0.05, 0.10, and 0.20. INKA rankings were moderately stable but more threshold-dependent, with off-diagonal overlaps ranging from 0.30 to 0.90.

Overall, SCZ top-ranked LIKA kinases were largely insensitive to the first-stage FDR threshold. INKA showed greater threshold dependence, particularly between stringent and relaxed thresholds, but the default FDR 0.05 ranking remained close to neighboring settings, with 90% top-10 overlap against both FDR 0.01 and FDR 0.10.

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